

Problem 1.6

Using the formula of the previous problem, in how many years will \$500 accumulate to \$630 if the annual interest rate is 7.8%?

Solution

The simple interest formula is

$$A = P(1 + it).$$

Substituting the given values $A = 630$, $P = 500$, and $i = 7.8\% = 0.078$,

$$630 = 500(1 + 0.078t).$$

Divide both sides by 500:

$$\frac{630}{500} = 1 + 0.078t.$$

Then,

$$\frac{630}{500} - 1 = 0.078t.$$

Since

$$\frac{630}{500} = 1.26,$$

we get

$$1.26 - 1 = 0.078t,$$

so

$$0.26 = 0.078t.$$

Therefore,

$$t = \frac{0.26}{0.078} = 3.333 \dots$$

Hence, the money will accumulate in approximately

$$\boxed{3.33 \text{ years}}.$$